SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 20%

QUESTIONS:3 **Total Marks:30**

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift

   ii. D2 must work before D3 (Morning < Afternoon < Night)

   iii. Only one doctor per shift

   iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.
   The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

   Constraints:

   - The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:
a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I **Abdul Razak Shaik _192524358____** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Abdul Razak Shaik**

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

1.Backtracking Search – Doctor Shift Assignment

Given

Doctors:

- D1
- D2
- D3

Shifts (Order):

- Morning
- Afternoon
- Night

Constraints

1. D1 cannot work **Night**.
2. D2 must work **before** D3 (Morning < Afternoon < Night).
3. Only **one doctor** per shift.
4. D3 cannot work **Morning**.
5. Every doctor must be assigned **exactly one shift**.

Step 1: Start with Empty Assignment

Doctor	**Shift**
D1	—
D2	—
D3	—

Step 2: Assigning D1

Possible shifts for D1:

- Morning
- Afternoon
- Night (Constraint 1) [not possible]

So we consider **Morning**.

Doctor	Shift
D1	Morning
D2	—
D3	—

Constraint Check:

- D1 not on Night
- No duplicate shift

Step 3: Assigning D2

Available shifts:

- Afternoon
- Night

So we considering **Afternoon**.

Doctor	Shift
D1	Morning
D2	Afternoon
D3	—

Constraint Check:

- One doctor per shift
- D2 before D3 (not checked until D3 assigned)

Step 4: Assigning D3

Remaining shift:

- Night

Check constraints:

- D3 cannot work Morning
- D2 works before D3
    - Afternoon < Night
- One doctor per shift

Assignment:

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

All constraints satisfied.

Backtracking Search Tree

```
Start
│
├── D1 = Morning
│    │
│    ├── D2 = Afternoon
│    │     │
│    │     └── D3 = Night
│    │
│    └── D2 = Night
│          │
│          └── D3 = Afternoon
│           (D2 is after D3, it violates Constraint ii)
│
└── D1 = Afternoon
     │
     ├── D2 = Morning
     │     │
     │     └── D3 = Night
     │
     └── D2 = Night
           │
           └── D3 = Morning
          (D3 cannot work Morning)
```

Intermediate Constraint Checks

Step	Assignment	Constraint Check	Result
1	D1 = Morning	D1 not Night	Yes
2	D2 = Afternoon	Unique shift	Yes
3	D3 = Night	D3 not Morning	Yes
4	D2 before D3	Afternoon < Night	Yes
5	All doctors assigned	Exactly one shift each	yes

Final Valid Schedule

Shift	Doctor
Morning	**D1**
Afternoon	**D2**
Night	**D3**

Final Answer:

Morning → D1,
Afternoon → D2,
Night → D3

2. Robot Path Finding Using Breadth-First Search (BFS)

Problem

A robot has to move in a **5 × 5 grid** from the **Start (S)** position to the **Goal (G)** position.
Constraints

- Movement allowed: **Up, Down, Left, Right**
- Cost of each move = **1**
- Robot cannot move through obstacles.
- Use **Breadth-First Search (BFS)** to find the shortest path.

Row/Column	1	2	3	4	5
1	S	Free	Free	X	Free
2	Free	X	Free	X	Free
3	Free	X	Free	Free	Free
4	Free	Free	X	X	Free
5	Free	Free	Free	Free	G

Legend

- **S** = Start
- **G** = Goal
- **X** = Obstacle
- **Free**= open path

Step-by-Step BFS Search

Step 1
Current Node: S (1,1)
Possible moves:

- Right → (1,2)
- Down → (2,1)

Queue: (1,2), (2,1)

Step 2
start: (1,2)
Possible move:

- Right → (1,3)

(2,2) is blocked.
Queue: (2,1), (1,3)

Step 3
start: (2,1)
Possible move:

- Down → (3,1)

Queue: (1,3), (3,1)

Step 4
start : (1,3)
Possible move:

- Down → (2,3)

Queue: (3,1), (2,3)

Step 5
start: (3,1)
Possible move:

- Down → (4,1)

Queue: (2,3), (4,1)

Step 6
start: (2,3)
Possible move:

- Down → (3,3)

Queue: (4,1), (3,3)

Step 7
start (4,1)
Possible moves:

- Right → (4,2)
- Down → (5,1)

Queue: (3,3), (4,2), (5,1)

Step 8
start (3,3)
Possible move:

- Right → (3,4)

Queue: (4,2), (5,1), (3,4)

Step 9
start: (4,2)
No new valid node.
Queue: (5,1), (3,4)

Step 10
start: (5,1)
Possible move:

- Right → (5,2)

Queue: (3,4), (5,2)

Step 11
start: (3,4)
Possible move:

- Right → (3,5)

Queue: (5,2), (3,5)

Step 12
start: (5,2)
Possible move:

- Right → (5,3)

Queue: (3,5), (5,3)

Step 13
start: (3,5)

Possible move:

- Down → (4,5)

Queue: (5,3), (4,5)

Step 14
start: (5,3)
Possible move:

- Right → (5,4)

Queue: (4,5), (5,4)

Step 15
start: (5,4)
Possible move:

- Right → **Goal (5,5)**

Goal reached.

BFS Search Table

Step	Node Expanded	New Nodes Added
1	(1,1)	(1,2), (2,1)
2	(1,2)	(1,3)
3	(2,1)	(3,1)
4	(1,3)	(2,3)
5	(3,1)	(4,1)
6	(2,3)	(3,3)
7	(4,1)	(4,2), (5,1)
8	(3,3)	(3,4)
9	(4,2)	None
10	(5,1)	(5,2)
11	(3,4)	(3,5)
12	(5,2)	(5,3)
13	(3,5)	(4,5)
14	(5,3)	(5,4)
15	(5,4)	Goal (5,5)

Cost Calculation

- Total number of moves = **8**
- Cost per move = **1**

Total Cost = 8

3.Autonomous Rescue Robot in a Disaster-Affected Building

a) Analysis of the Constraints

The rescue robot must reach the survivor while avoiding obstacles and minimizing the total cost.

Constraint	Effect on Robot's Decision-Making
Robot can move Up, Down, Left, Right	The robot explores only four possible directions from its current position.
Obstacles cannot be traversed	The robot must avoid blocked cells and search for an alternative route.
Limited sensing capability	The robot can observe only adjacent cells, so it has incomplete knowledge of the environment.
Movement cost = 1	Every movement increases the total path cost by 1.
Risky zone cost = +2	The robot should avoid risky areas whenever a safer path exists.
Dynamic knowledge update	As the robot discovers new obstacles or risky zones, it updates its map and replans its path.

b) Solution Approach (Using Uniform Cost Search with Online Search)

Step 1: Start Position

- The robot starts from **S**.
- Initially, only adjacent cells are known.

Step 2: Observe the Environment

- Sense the neighbouring cells.
- Identify free paths, obstacles, and risky zones.

Step 3: Calculate Cost

- Normal movement cost = **1**
- Risky cell cost = **1 + 2 = 3**

Step 4: Expand the Lowest-Cost Node

- Select the path with the minimum accumulated cost.
- Ignore blocked cells.

Step 5: Update Knowledge

- If a new obstacle or risky area is detected, update the map.
- Recalculate the best path.

Step 6: Continue Until Goal

- Repeat the process until the survivor (**G**) is reached.

Step 7: Rescue Survivor

- Reach the goal with the minimum possible total cost.

c) Application of AI Concepts

1. Intelligent Agent

The rescue robot acts as an **Intelligent Agent** because it:

- Perceives the environment using sensors.
- Processes the available information.
- Selects the best action.
- Updates its knowledge while moving.
- Reaches the survivor efficiently.

2. Rationality

The robot behaves **rationally** because it:

- Chooses the least-cost path.
- Avoids obstacles.
- Minimizes travel through risky zones.
- Continuously updates its decisions when new information is available.

3. Environment Type

The rescue environment is:

Property	**Type**
Observability	Partially Observable
Agents	Single Agent
Nature	Dynamic
Outcomes	Deterministic
State Space	Sequential
Knowledge	Unknown at the beginning

d) Justification of the Chosen Approach

The **Uniform Cost Search (UCS)** combined with an **Online Search Agent** is the most suitable approach because:

- It always finds the **minimum-cost path**.
- It considers different movement costs (normal and risky zones).
- It avoids unnecessary travel through high-cost areas.
- It can dynamically update its path as new information becomes available.
- It is well suited for **partially observable** and **dynamic** environments, where the complete map is not known initially.

Flowchart of the Solution

Start

Sense Adjacent Cells

Identify Free, Blocked & Risky Cells

Calculate Path Cost

Choose Lowest-Cost Path (UCS)

Move to Next Cell

Update Knowledge

Reached Goal?

No → Repeat

Yes → Rescue Survivor → End